

Homework 3

Student Name

Discrete Mathematics

Problem 1. *Prove that for every integer n , the number $n^3 - n$ is divisible by 6.*

Solution. Factor $n^3 - n = (n - 1)n(n + 1)$. Among three consecutive integers one is divisible by 3 and at least one is even, so the product is divisible by $2 \cdot 3 = 6$. $\square$

Problem 2. *Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = 3x - 7$.*

(a) *Show that f is injective.*

(b) *Show that f is surjective.*

(c) *Find f^{-1} .*

Solution. (a) If $f(x) = f(y)$ then $3x - 7 = 3y - 7$, so $x = y$.

(b) Given $y \in \mathbb{R}$, the number $x = (y + 7)/3$ satisfies $f(x) = y$.

(c) From part (b), $f^{-1}(y) = \frac{y + 7}{3}$. $\square$

Problem 3 (Pigeonhole). *Show that among any $n + 1$ integers there are two whose difference is divisible by n .*

Solution. There are only n residues modulo n , so by the pigeonhole principle two of the $n + 1$ integers, say a and b , share a residue. Then

$$a \equiv b \pmod{n}, \quad \text{that is, } n \mid a - b. \quad \square$$

Problem 4. *Decide whether each statement is true or false, and justify your answer.*

1. $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z} : x + y = 0$.

2. $\exists y \in \mathbb{Z}, \forall x \in \mathbb{Z} : x + y = 0$.

Solution. The first statement is true: take $y = -x$. The second is false, since no single y satisfies both $0 + y = 0$ and $1 + y = 0$. $\square$